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FULL STACK

- React, Next.js, TypeScript
- Node.js, Express, FastAPI
- REST APIs, webhooks, auth
- Dashboards and portals

AI & DATA

- Claude / OpenAI / local LLMs
- Agent tools, RAG, memory
- Oracle APEX, SQL, Postgres
- MongoDB, Prisma, Supabase

MOBILE & CLOUD

- Flutter / Dart / iOS / Android
- Windows and macOS apps
- AWS/cloud services
- CI/CD, Docker, observability

EMBEDDED

- C / C++ / microcontrollers
- BLE, CAN, OBD2, UART
- Firmware and bootloaders
- Telemetry and diagnostics

DOMAIN

- Industrial machine software
- Automotive tuning platforms
- Commerce and operations
- Scientific / regulated systems

Matthew Phillips

Full-Stack Software Engineer / Engineering Lead

AI | Mobile | Cloud | Embedded | Industrial & Automotive Systems

github.com/Stericallyhindered/PORTFOLIO

Summary

Hands-on software engineer and engineering lead with more than 11 years of experience building complete systems across full-stack web, AI/agent workflows, mobile and desktop applications, cloud infrastructure, databases, embedded firmware, industrial machinery, automotive software, ecommerce, analytics, and internal business operations. Experienced with React, Next.js, TypeScript, Node.js, Python, FastAPI, Flutter/Dart, C/C++, C#, Java, SQL, Oracle APEX, PostgreSQL, MongoDB, AWS/cloud services, Docker, CI/CD, and production operations.

I build the connective tissue between products, users, devices, data, APIs, operations, and support teams. My work includes customer portals, admin dashboards, API integrations, AI support agents, LLM workstation tools, ecommerce platforms, CRM/order systems, warranty workflows, inventory and shipping automation, mobile apps, firmware update flows, telemetry, diagnostics, validation tools, background jobs, webhooks, audit trails, and deployment pipelines.

My background is intentionally broad: industrial CAD/CAM and machine-control software, cross-platform Flutter applications, embedded C/C++ firmware, bootloaders, Python test automation, BLE/CAN/OBD2/UART communication, high-volume vehicle telemetry, cloud APIs, database architecture, practical AI systems, Oracle APEX business applications, ecommerce operations, MP Analytical custom scientific software, HPLC/mass spec workflows, ISO lab practices, and regulated analytical testing. This range lets me debug across application, infrastructure, data, device, firmware, laboratory, and human-process boundaries.

Full-Stack, Mobile, AI, and Embedded Profile

- Own software systems from workflow discovery and technical strategy through architecture, implementation, CI/CD, monitoring, documentation, production support, and handoff.
- Refactor producer- and junior-developer prototypes into maintainable software by defining typed contracts, separating concerns, replacing temporary storage, and adding authentication, authorization, validation, structured logging, tests, deployment configuration, and recovery paths.
- Build directly with product managers, producers, designers, engineers, technicians, support teams, sales, shipping, manufacturing, and business operations; comfortable holding several small-to-medium workstreams in ambiguous environments.
- Led five developers at Burger Motorsports and three junior developers at Stealth while remaining a hands-on architect and primary implementer across automation, full-stack, mobile, firmware, data, AI, and production operations.
- Strong fit for full-stack product engineering, mobile applications, AI systems, internal platforms, commerce/admin tooling, hardware-connected software, data pipelines, developer enablement, and production systems where reliability matters.

Core Strengths

- Production workflow automation: advanced Airtable bases, scripting, REST API usage, webhooks, linked records, cross-base and cross-project views, Dev/production validation, scheduled and event-driven automations, idempotent processing, reconciliation, and clear failure/retry states.
- Collaboration and enterprise integrations: Slack, Signal, WhatsApp, Google Workspace, Jira, Glean-style knowledge workflows, GitHub, Twilio, email providers, webhooks, OAuth/JWT-style authentication, rate-limit handling, and multi-channel notification and escalation.
- LLM Workstation Hub: channel adapters that normalize Slack, Signal, and WhatsApp events into shared message contracts; route requests through local or hosted models and approved tools; retain authorized history, operational context, and feedback; and return responses or escalate work to the originating channel.
- CI/CD and infrastructure: GitHub-based DevOps workflows, automated build/test/deploy pipelines, Docker and Docker Compose, cloud deployments, environment and secret management, infrastructure-as-code practices, release validation, observability, and production troubleshooting.
- Full-stack product engineering: React, Next.js, TypeScript, JavaScript, Node.js, Express, Python, FastAPI, REST APIs, WebSockets, auth, permissions, dashboards, admin tools, customer portals, and deployment documentation.
- Backend and integration work: API integrations, webhooks, provider adapters, notification flows, Stripe/payment workflows, shipping workflows, ticketing, logging, audit trails, role-aware state machines, structured operational data, and Oracle APEX-connected business databases.
- Oracle APEX and database application engineering: Oracle APEX, SQL, PL/SQL-style workflow thinking, customer/warranty/automation databases, reporting, production support, documentation, testing, and enhancements.
- AI systems: Claude/Anthropic and OpenAI-style integrations, AI support agents, chat workflows, document/context ingestion, ticket triage, warranty/claims intake, onboarding, knowledge-base lookup, analytics, escalation paths, and human handoff flows.
- Internal tools and workflow platforms: CRM/order systems, developer/support tooling, operational dashboards, approval workflows, customer records, machine/device records, sales/dealer portals, affiliate workflows, field-service tools, and support history.
- Industrial machine and CAD/CAM software: full-stack operator and technician interfaces for fiber laser welders and cutters, plasma CNC equipment, press brakes, benders, and related machine tools; CAD/CAM workflows, job and machine configuration, diagnostics, telemetry, field support, and remote service tooling.
- Mobile and hardware-connected software: Flutter/Dart applications across Windows, macOS, iOS, and Android; BLE, UART/serial, CAN bus, OBD2, TCP/IP, device discovery, live telemetry, diagnostics, datalogging, settings, calibration, and application-to-device control.
- Firmware and embedded systems: C/C++, microcontrollers and embedded development platforms, ESP32/ESP-IDF, Arduino, Raspberry Pi-class systems, device drivers, interrupts, state machines, watchdogs, persistent configuration, bootloaders, remote firmware delivery, HEX parsing, block erase/write, checksum validation, and recovery behavior.
- Firmware validation and test engineering: Python automated test scripts, regression testing, protocol and packet validation, hardware-in-the-loop and bench testing, failure reproduction, instrumentation, logging, CI-aware test workflows, release validation, and root-cause analysis.
- Vehicle, IoT, and telematics systems: ECU/tuning workflows, cellular/GPS/connected-device concepts, BLE/CAN/OBD2 telemetry, raw frame capture, signal decoding, thousands of logged parameters, derived metrics, fuel/sensor controllers, and embedded Linux/networking concepts.
- Scientific and regulated background: MP Analytical owner/operator, biochemistry, HPLC, mass spectrometry, analytical testing software, ISO lab standards, medical/technical device-adjacent code, data integrity, traceable workflows, documentation, validation, and quality-focused change control.
- Working style: ask scope questions early, understand the user, keep changes readable, document decisions, review AI-generated code critically, communicate before going dark, and own production behavior.

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Professional Experience

Stealth Machine Tools / Stealth Batteries

Software Engineer / AI Engineer / Technical Lead

May 2025 - Present

- Lead and mentor three junior developers while building full-stack software, AI integrations, CRM systems, support automation, dashboards, field/support apps, and internal tools for Stealth Machine Tools, Stealth Batteries, and related businesses.
- Build and maintain advanced Airtable applications for production, product, support, inventory, customer, machine, and cross-project workflows using linked data models, scripts, REST APIs, webhooks, scheduled and event-driven automations, validation, text/data checks, approvals, and synchronized operational views.
- Built the Stealth LLM Workstation Hub integrating Slack, Signal, and WhatsApp through channel adapters, normalized message contracts, local and hosted LLM routing, approved tool execution, persisted conversation and business context, feedback capture, automated responses, and human escalation back to the originating channel.
- Own GitHub-centered DevOps and CI/CD workflows for Stealth software, including automated build, test, and deployment stages, containerized services, environment and secret management, cloud deployments, release validation, logging, monitoring, and production incident troubleshooting.
- Regularly take producer- and junior-developer prototypes from proof of concept to production by restructuring modules, defining typed contracts, replacing temporary data paths, adding permissions, validation, tests, observability, deployment configuration, and operational documentation.
- Work directly across product management, production, design, engineering, sales, support, shipping, manufacturing, and field operations to map workflows, prioritize automation, define requirements, and deliver zero-to-one tools that remove coordination overhead.
- Architected and developed full-stack CAD/CAM, operator, configuration, diagnostics, and support interfaces for fiber laser welders and cutters, plasma CNC machines, press brakes, benders, and related industrial equipment.
- Built cross-platform Flutter/Dart machine applications for Windows, macOS, iOS, and Android, connecting user interfaces and cloud services to machine settings, telemetry, logs, diagnostics, firmware, customer records, and remote-support workflows.
- Worked across embedded controllers, microcontrollers, sensors, machine communications, firmware behavior, and application-to-device integration; developed Python tooling for automated validation, repeatable diagnostics, regression testing, log analysis, and production troubleshooting.
- Supported firmware and hardware testing with UART/serial, TCP/IP, BLE, embedded Linux and microcontroller development platforms, and laboratory tools including oscilloscopes, multimeters, logic analyzers, and bench power supplies.
- Built AI-assisted support and internal workflows combining chat, onboarding, warranty intake, machine/product context retrieval, versioned prompts, tool/API calls, structured validation, confidence thresholds, retained history, analytics, escalation, and role-aware human review.
- Built Oracle APEX-backed customer, warranty, automation, and support databases for Stealth Machine Tools and Burger Motorsports, turning business requirements into database applications instead of manual spreadsheets and inbox tracking.
- Built Stealth Batteries commerce/admin platform using Next.js, React, TypeScript, Payload CMS, Postgres, Stripe, webhooks, email providers, shipping APIs, dealer portals, affiliate workflows, sales-rep dashboards, order management, warranty/support workflows, and custom admin tools.
- Built Stealth Machine backend tooling with Next.js/TypeScript/Prisma patterns around auth, customers, machines, tickets, training/support materials, AI configuration, analytics, and admin workflows.
- Built cross-platform Flutter field/support tooling for machine support and training workflows, including API integration, local storage, auth helpers, camera/speech/PDF support, QR-style workflows, charts, and field-service user experience.
- Built CRM/order/customer tracking systems with Node/Express, SQLite, JWT auth, role permissions, admin approvals, customer portals, document intake, real-time updates, Twilio/Nodemailer notifications, audit trails, deployment notes, and Oracle APEX-connected operational records.
- Connected portals, admin actions, approvals, document intake, notifications, reporting, support history, warranty claims, inventory, shipping/returns, and customer records into structured workflows that reduced manual operational load.

Burger Motorsports

Lead Developer - Automotive Software, Mobile Apps, Hardware Integration, Support Platforms

2019 - 2025

- Served as principal vehicle software engineer and lead developer across the complete JB4 product stack: embedded firmware, bootloaders, tuning logic, connected hardware, cross-platform user interfaces, backend/support systems, databases, and production validation for an ecosystem used by more than 500,000 vehicles worldwide.
- Owned zero-to-one product, producer, design, architecture, implementation, release, and production-support responsibilities while translating requirements from ownership, tuners, support staff, hardware engineering, and customers into dependable software.
- Built private internal dashboards, administrative portals, API services, and automation for vehicle-owner records, support queues, telemetry and datalog analysis, firmware/version context, tuning history, product operations, and high-volume web and data workflows.
- Coordinated work across software, hardware, firmware-adjacent requirements, vehicle platform behavior, support needs, customer usability, release risk, and product documentation.
- Built the full-stack interface around JB4 tuning hardware, including Flutter/Dart applications for iOS, Android, Windows, and macOS; BLE/serial device communication; cloud and support services; Oracle APEX operational databases; and customer, tuner, and developer interfaces.
- Developed and supported embedded C/C++ firmware and microcontroller integrations for JB4 devices, CANFlex controllers, standalone GDI drivers, fuel injector controllers, sensor modules, and related automotive hardware.
- Built vehicle tuning, calibration, diagnostics, and datalogging systems capable of collecting, decoding, visualizing, and exporting thousands of parameters across many vehicle platforms, ECUs, firmware versions, maps, and hardware configurations.
- Implemented remote application-to-device firmware delivery including firmware retrieval and manual HEX loading, bootloader entry, erase/write commands, block flashing, checksums, version validation, progress state, recovery handling, bootloader exit, and user-facing safety controls.
- Built automated and bench validation workflows using Python and engineering tools to test firmware, protocols, telemetry, connection recovery, regression behavior, and hardware/software interaction before release.
- Built command/protocol logic for JB4 hardware including settings retrieval, settings save, logging start/pause, map selection, E85/flex fuel configuration, WMI/meth settings, boost/fuel tables, AFR/meth/fuel pressure telemetry, and developer packet inspection.
- Developed CANFlex/NewCANFlex mobile hardware software for fuel and sensor control: BLE device communication, ethanol content, fuel temperature, fuel pressure, RPM, speed, water/oil temperature, gear and torque surfaces, CAN bus output configuration, analog output modes, pressure sensor modes, calibration controls, firmware/version display, live gauges, logs, and ECU/JB4 integration paths.
- Built CRM-backed support tooling that replaced manual email-based log review with structured customer submissions, vehicle log intake, AI-assisted parsing, diagnostic/tuning guidance, support history, escalation paths, and Oracle APEX-backed operational records.
- Helped reduce operational support burden by creating systems that allowed technical support workflows, product context, vehicle logs, and customer history to live in structured tools instead of inboxes and spreadsheets.
- Worked close to embedded systems, custom hardware, firmware, and vehicle communication protocols, translating raw device/vehicle data into reliable UI state and supportable diagnostic workflows.
- Built across desktop and mobile hardware interfaces, including legacy C# WinForms/serial/BLE support code and newer Flutter mobile tooling.
- Operated in a domain where incorrect parsing, settings, firmware assumptions, or support guidance could affect real vehicles, requiring careful validation, documentation, ownership, and code review.
- Completed the major software initiatives ahead of schedule, documented and transitioned the systems, and moved on once the core work had been delivered and the role naturally wound down.

MP Analytical

Owner / Scientific Software and Analytical Testing Systems

2015 - 2021

- Owned and operated MP Analytical while building custom software and technical workflows for analytical testing, laboratory operations, data review, reporting, traceability, and scientific process automation.
- Worked heavily with HPLC, mass spectrometry, analytical testing practices, ISO lab standards, regulated documentation, data integrity, instrument-driven workflows, method context, result review, and quality-focused change control.
- Built custom tools that helped turn raw analytical data, instrument output, sample context, batch/run information, reports, and customer or internal requirements into structured workflows technical users could trust.
- Developed software around scientific data handling, calculations, validation checks, reporting support, audit-friendly records, exception review, repeatable procedures, and practical troubleshooting in lab environments.
- Brought software engineering into analytical operations by reducing spreadsheet/manual-process risk, improving traceability, documenting assumptions, and building tools around how scientists and technicians actually work.

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Additional Experience

Bayer

Technical Intern / Analytical Testing Software and Embedded Lab Systems

2015 - 2018

- Started as an internship that turned into a professional engineering role, working inside analytical and medical-adjacent lab environments where process control, documentation, validation, and traceability mattered every day.
- Learned and supported analytical testing equipment, scientific measurement workflows, instrument behavior, sample/result handling, troubleshooting, and strict laboratory protocols.
- Wrote and supported embedded firmware and custom software around analytical testing equipment, helping connect lab instruments, technical workflows, data handling, and repeatable operating procedures.
- Developed a deep working understanding of ISO lab practices, regulated medical/testing environments, quality systems, change control, data integrity, and the discipline required to navigate highly controlled analytical processes.
- Built the foundation for how I still engineer today: measure carefully, validate assumptions, document decisions, respect protocol, and treat small software or workflow mistakes as real production risk.

Selected Project Highlights

Tech Support AI Backend

Production AI support backend that turned scattered customer conversations into structured operational workflows. Built Claude/LLM integration, JWT auth, MongoDB/Mongoose data models, machine/customer records, support tickets, document ingestion, analytics, notifications, and API routes so support could track real issues instead of losing context in inboxes.

Stealth AI Support System

Agentic customer and internal support platform for complex industrial products. Connected Claude-style chat, real-time support workflows, admin dashboards, ticketing, analytics, conversation history, onboarding, warranty/claims handling, escalation rules, and human handoff so customers could get guided technical help without waiting for manual triage.

Stealth Batteries Commerce/Admin Platform

Full production commerce and operations platform for a physical-product business. Connected ecommerce, products, dealers, affiliates, sales reps, orders, shipping, support, warranty, dashboards, and Payload/Postgres admin tooling, replacing manual email/status tracking with a system the business could actually run from.

Stealth Machine Backend and Field App

Industrial machine support stack combining a Next.js/Prisma backend with a Flutter field app. Built machine records, customer context, support tickets, training materials, AI configuration, analytics, local storage, camera/speech/PDF helpers, QR-style workflows, charts, and field-service UX so technicians had the machine history and tools in hand.

Industrial Machine CAD/CAM and Control Software

End-to-end CAD/CAM, operator, configuration, diagnostics, telemetry, and remote-support software for fiber laser welders and cutters, plasma CNC machines, press brakes, benders, and related manufacturing equipment. Spanned Flutter/Dart mobile/desktop interfaces, backend services, databases, machine communications, embedded controllers, firmware workflows, and field validation for equipment used in real manufacturing.

Customer Tracking CRM

Custom CRM and order operations platform for real business throughput. Built company/contact/order workflows, approvals, customer portal views, document uploads, OCR-style processing, real-time updates, notifications, audit trails, role-aware state, and Oracle APEX-connected database operations to make sales, support, and fulfillment traceable.

GEO Command Center Platform

Full-stack AI/search visibility SaaS architecture with Next.js, Supabase/Postgres, Drizzle, FastAPI, background workers, tests, provider/report pipelines, audits, citation tracking, multi-tenant dashboards, and analytics. Built as a serious SaaS foundation, not a demo, with data modeling and operational reporting from the start.

JB4Pro Vehicle Tuning Platform

End-to-end embedded and full-stack platform for JB4 tuning hardware used across a massive active vehicle ecosystem. Built C/C++ firmware workflows, microcontroller integrations, bootloaders, remote app-to-device firmware delivery, BLE/serial/CAN communication, Flutter apps across mobile and desktop, calibration tools, live telemetry, thousands of logged parameters, map/settings control, E85/flex fuel and WMI/meth workflows, diagnostics, cloud/support systems, and release validation.

CANFlex Fuel and Sensor Controller

Mobile and hardware-connected software for a fuel injector, E85 sensor, and fuel-pressure control ecosystem. Built BLE communication, ethanol/fuel-temperature/fuel-pressure gauges, CAN bus and analog output configuration, pressure sensor modes, calibration, logs, firmware/version handling, and ECU/JB4 integration paths for real vehicle tuning use.

TurboLamik AWD / Transmission Telemetry

Vehicle and embedded telemetry system for transmission and AWD control research. Built ESP-IDF-oriented C scaffolding, BMW E90 CAN signal decoding, TurboLamik TCU frame decoding, raw CAN capture, health/watchdog state, derived drivetrain metrics, shadow AWD request logic, BLE telemetry, Flutter dashboard, raw frame monitor, decoded signal monitor, and saved log/export workflows.

STR8Tune ECU Calibration Editor

ECU calibration editor built around the workflows tuners actually need: React/TypeScript UI, XDF/BIN parsing concepts, editable tables/maps, hex and calibration views, tuning assistant flows, comparison tools, 2D/3D visualization, and export-style workflows for making binary calibration data understandable.

AmpGen Configurator

Next.js/React configurator that turned electrical/power-system planning into an interactive product workflow. Built draggable layout zones and TypeScript UI logic so customers and operations could reason about a physical system visually instead of through back-and-forth manual notes.

Industrial Commerce and Admin Systems

Commerce and admin systems for industrial and specialty-product companies where inventory, payment, shipping, and support had to line up. Built inventory services, manual order flows, checkout/payment logic, shipping dashboards, customer/admin workflows, migrations, reusable components, API documentation, and handoff notes.

Windows and Network Automation

Operational automation for Windows and network-heavy environments. Built Python/PowerShell local APIs and control tooling around hotspot/network workflows, the kind of unglamorous infrastructure code that keeps real deployments usable when people are under pressure.

Legacy Hardware Desktop Interfaces

Maintained and modernized legacy C# WinForms hardware interfaces where reliability mattered more than fashion. Worked around serial port detection, Bluetooth/BLE support, Windows UI surfaces, device discovery, and tuning-tool ergonomics so older production tooling stayed useful while newer platforms came online.

Technologies

Languages: JavaScript, TypeScript, Python, SQL, Dart, C#, C++, C, Java, HTML, CSS, Bash, PowerShell, VBA

Frontend: React, Next.js, Tailwind CSS, Radix/shadcn-style UI, dashboard UI, admin panels, customer portals, configuration interfaces, charts/gauges, table/map editors, 2D/3D visualization concepts

Backend: Node.js, Express, FastAPI, REST APIs, WebSockets, JWT, OAuth concepts, role permissions, audit logging, provider adapters, webhooks, background jobs, notification workflows

Automation / Data: Airtable scripting, Airtable REST APIs and automations, linked records, cross-base/project synchronization, Oracle APEX, SQL, PL/SQL-style workflows, Postgres, SQLite, MongoDB/Mongoose, Supabase, Drizzle, Prisma, Redis/BullMQ queues, Payload CMS, migrations, validation, audit trails, and reconciliation

AI / Integrations: Slack, Signal, WhatsApp, Google Workspace, Jira, Glean-style knowledge integration, Claude/Anthropic, OpenAI and local LLMs, agent tools, prompt/context workflows, retained history, document ingestion, OCR, ticket triage, Stripe, ShipStation/shipping APIs, Resend/Nodemailer, Twilio, and webhooks

Industrial / CAD-CAM: fiber laser welders and cutters, plasma CNC systems, press brakes, benders, operator HMIs, CAD/CAM workflows, machine configuration, diagnostics, telemetry, remote support, and field-service software

Mobile / Hardware: Flutter, Dart, iOS, Android, Windows, macOS, BLE, UART/serial, CAN bus, OBD2, TCP/IP, cellular/GPS/telematics concepts, raw frame capture, diagnostics, high-volume datalogging, gauges, calibration, and application-to-device control

Embedded / Firmware: C, C++, microcontrollers, ESP32, ESP-IDF, Arduino, Raspberry Pi, embedded Linux, GPIO, PWM, interrupts, drivers, watchdogs, state machines, persistent configuration, bootloaders, remote firmware updates, HEX parsing, erase/write, checksums, and recovery flows

Validation / Test: Python automation, firmware validation, regression testing, protocol and packet tests, hardware-in-the-loop and bench testing, CI pipelines, automated test frameworks, logging, failure reproduction, root-cause analysis, oscilloscopes, logic analyzers, multimeters, power supplies, and serial/CAN tools

Vehicle / Telematics: ECU calibration and tuning, JB4, CANFlex, GDI drivers, CAN RX dispatch, signal stores, thousands of telemetry parameters, BMW E90 decoding, TurboLamik TCU decoding, MaxxECU CAN integration, fuel/sensor controller workflows, and connected-device diagnostics

DevOps / Ops: GitHub, CI/CD pipelines, Docker and Docker Compose, AWS/GCP/Azure, infrastructure-as-code workflows, automated tests, environment and secret management, logging, monitoring, incident response, C# WinForms, Python/PowerShell automation, and technical documentation

Education

Biochemistry and Software Engineering coursework, 2012 - 2015

Studied biochemistry with a software engineering focus while preparing to build analytical testing and technical workflow software.

How I Work

I am comfortable with "here is the problem, go figure it out" work. I ask scope questions early, especially around users, safety, diagnostics, edge cases, ownership, and production risk. I prefer simple architecture, clear data flow, validation before control output, and code that another engineer can safely change. I communicate before going dark, leave notes/change logs, and document enough context for the next engineer or AI-assisted workflow to understand the system later.